STATS 507 Proposal: Attention is also Needed for Time Series

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1 Overview

This project addresses time series forecasting, a critical task in fields like finance, weather prediction, and resource management, enabling better decisions based on future trends. Traditional methods often fall short in capturing complex, multivariate dependencies, while newer transformer-based models like Triformer (1) utilize advanced mechanisms such as attention and decomposition to tackle these challenges. However, their effectiveness in long-term forecasting remains uncertain, necessitating further investigation into their robustness and adaptability to diverse sequence lengths.

2 Prior Work

The vanilla transformer, introduced in the landmark "Attention is All You Need" paper (2), forms the foundation of all transformer models. To highlight the limitations of traditional time series approaches, the classic statistical ARIMA/ETS methods discussed by Hyndman (3) are also reviewed.

Transformer adaptations for time series data, such as Triformer(1), demonstrate superior performance over traditional RNN/CNN models on specific datasets. These works provide a foundation for evaluating time series forecasting, offering baseline methods and experimental designs that will inform our more comprehensive investigations.

The variant that will be used was introduced by Cirstea et al. (1), the Triformer. The paper is currently under review.

3 Preliminary Results

3.1 Data

We will conduct initial evaluations using the 2019-2024 US Stock Market Data from Kaggle, introduced by Saket (4), to assess the performance of Transformers on time series forecasting. This dataset provides a comprehensive view of market behavior over five years, covering price movements and trading volumes across diverse sectors. It includes energy commodities, metals, cryptocurrencies, and major stock indices and companies. This data offers valuable insights for analyzing trends in global markets.

3.2 Models

This project will use a transformer model for time series forecasting, based on the attention mechanism(2). We will also implement and compare this model with Triformer(1). Hugging Face resources will be utilized to deploy both models. A smaller dataset will be used for efficient training and evaluation. The following is an image explaining the architecture of a vanilla transformer that I found intuitive to understand from Medium.

4 Project Deliverables

The project deliverables include training a vanilla transformer model, presenting its performance metrics and predictions, and deploying a pretrained Triformer to compare its results. The final outcomes will highlight the performance metrics and prediction results of both models, providing a detailed comparison.

4.1 Sub-Goals

Here are some milestones for the project:

1 Correctly deploy the Vanilla Transformer and is able to complete a short test training on a toy training set;

2 Correctly deploy Triformer and is able to complete short test training on a toy training set;

3 In case I have more time, I aim to complete training on the full dataset to obtain test and validation metrics;

4 Complete fine-tuning and optimize the model;

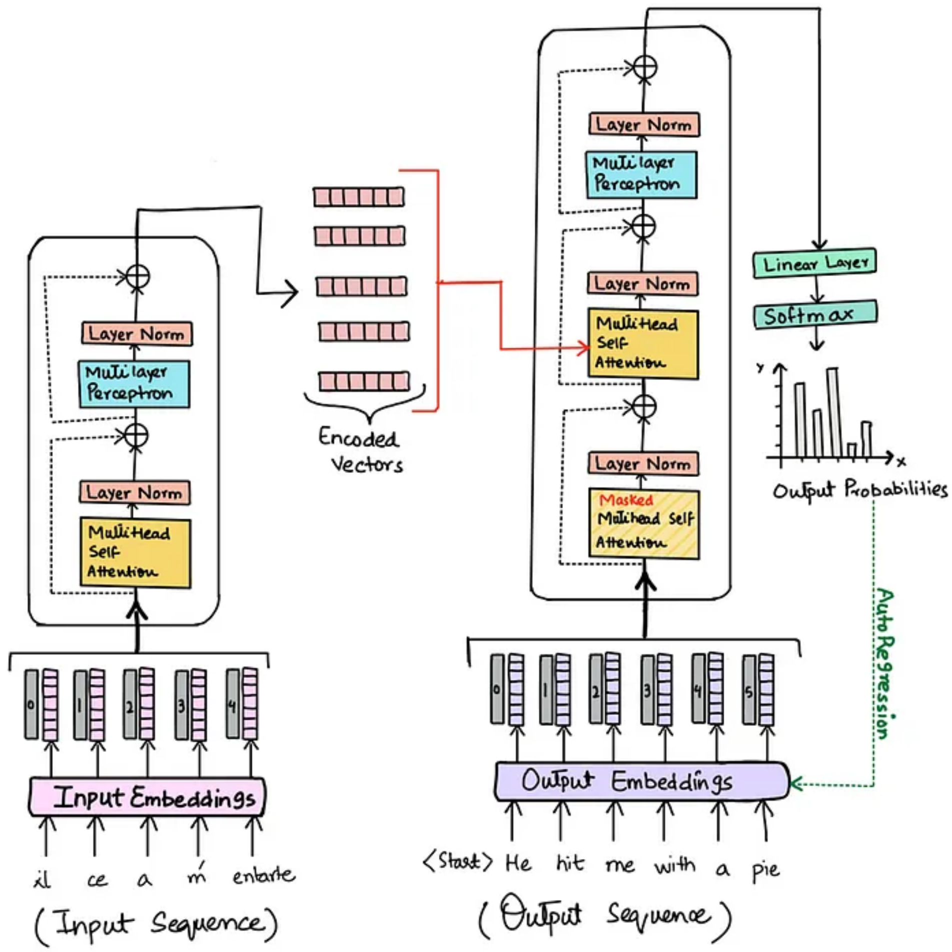

Figure 1: Vanilla Transformer Architecture

5 Compare the performance of the models with a test set, analyze the causes and identify potential fields of interest;

6 Apply the model in new areas and get better results than before with other models.

5 Timeline

Here is a rough timeline for the project by weeks:

1-2 Review prior works, EDA of dataset;

3-4 Deploy both models, finish trainings;

5 Test and compare results of the two models, write up the report.

References

[1] R.-G. Cirstea, C. Guo, B. Yang, T. Kieu, X. Dong, and S. Pan, "Triformer: Triangular, variable-specific attentions for long sequence multivariate time series forecasting," in *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, 2022.

[2] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[3] R. J. Hyndman and Y. Khandakar, "Automatic time series forecasting: The forecast package for R," *Journal of Statistical Software*, vol. 27, no. 3, pp. 1–22, 2008.

[4] S. Kumar, "2019-2024 US stock market data," 2024.